

Project Title:

Automated Text Classification: A Comparative Analysis of Machine Learning Models for Sports vs. Politics

Submitted by:

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1. Introduction

1.1 Background

Text classification is a fundamental task in Natural Language Processing (NLP) with applications ranging from spam detection to sentiment analysis and news categorization. As news aggregators process millions of articles daily, the ability to automatically tag content as "Sports" or "Politics" is critical for user experience and content recommendation systems.

1.2 Problem Statement

The objective of this assignment is to design a classifier that accepts a raw text document as input and predicts its category: **Sports** or **Politics**. The system must handle the nuances of natural language, such as synonymy (different words, same meaning) and polysemy (same word, different meanings). For example, the word "race" can refer to a political election or a track event, depending on the context.

1.3 Scope

This project compares three supervised learning techniques:

1. **Naive Bayes (NB):** A probabilistic baseline.
 2. **Logistic Regression (LR):** A discriminative linear model.
 3. **Support Vector Machines (SVM):** A max-margin classifier.
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2. Data Collection and Dataset Analysis

2.1 Data Source

We utilized the **20 Newsgroups** dataset, a benchmark dataset widely used in machine learning research. The dataset comprises approximately 20,000 newsgroup documents across 20 different categories.

For this specific binary classification task, we filtered the dataset to include only the following subsets:

- **Sports Class:** `rec.sport.baseball`, `rec.sport.hockey`
- **Politics Class:** `talk.politics.mideast`, `talk.politics.guns`, `talk.politics.misc`

2.2 Dataset Statistics

After filtering, the dataset was analyzed to ensure a sufficient number of samples for training.

- **Total Documents:** ~4,500
- **Sports Documents:** ~1,990 (44%)
- **Politics Documents:** ~2,510 (56%)

Analysis: The dataset is relatively balanced, though there is a slight skew toward Politics. This balance is crucial because severe class imbalance could lead the model to simply predict the majority class (Politics) for every input to achieve artificially high accuracy.

2.3 Data Cleaning

Raw text data contains noise that can confuse machine learning models. We applied the following preprocessing steps:

1. **Metadata Removal:** Headers, footers, and email quotes were stripped to ensure the model learns from the *content*, not the sender's email address.
2. **Lowercasing:** All text was converted to lowercase to treat "The" and "the" as the same token.
3. **Stop-Word Removal:** Common English words (e.g., "is," "and," "the") were removed using the Scikit-Learn stop-word list. These words carry high frequency but low semantic value.

3. Methodology and Feature Representation

3.1 Feature Extraction: TF-IDF

Computers cannot understand raw text; they require numerical input. We used **TF-IDF (Term Frequency-Inverse Document Frequency)** to convert text documents into numerical vectors.

- **Term Frequency (TF):** Measures how often a word appears in a specific document.
- **Inverse Document Frequency (IDF):** Measures how "unique" a word is across the entire dataset.
 - *Example:* The word "game" appears often in sports docs (High TF) but rarely in politics docs (High IDF), resulting in a high TF-IDF score.
 - *Example:* The word "said" appears in both (Low IDF), resulting in a low score.

3.2 N-Grams (Bi-grams)

We utilized an n-gram range of (1, 2). This means the vocabulary includes both single words ("tax") and two-word pairs ("tax cut"). This is vital for distinguishing context.

- *Uni-gram*: "Run" (Ambiguous)
 - *Bi-gram*: "Home run" (Sports) vs. "Run for office" (Politics).
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4. Machine Learning Techniques

4.1 Naive Bayes (MultinomialNB)

Naive Bayes is a probabilistic classifier based on Bayes' Theorem. It assumes that the presence of a particular feature (word) in a class is unrelated to the presence of any other feature.

- **Strengths**: Extremely fast training time and works well with high-dimensional text data.
- **Weaknesses**: The "independence assumption" is often false in human language (e.g., "White" is often followed by "House" in politics).

4.2 Logistic Regression

Logistic Regression is a linear model that learns a decision boundary between the two classes. It uses the **sigmoid function** to map predicted values to probabilities between 0 and 1.

- **Strengths**: Provides interpretability. We can examine the model weights to see which words contribute most to a "Sports" prediction (positive coefficients) vs. "Politics" (negative coefficients).

4.3 Support Vector Machines (SVM)

SVM constructs a hyperplane in a high-dimensional space to separate the two classes. It maximizes the "margin" (distance) between the hyperplane and the nearest data points of each class.

- **Configuration**: We used a **Linear Kernel** because text data is linearly separable in high dimensions.
 - **Strengths**: Robust against overfitting and generally provides the highest accuracy for text classification tasks.
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5. Quantitative Comparison & Results

We split the dataset into **80% Training** and **20% Testing** sets. The models were evaluated using Accuracy, Precision, Recall, and F1-Score.

5.1 Performance Metrics Table

Model	Accuracy	Precision (Sports)	Recall (Sports)	F1-Score (Weighted)
Naive Bayes	96.52%	0.98	0.96	0.96
Logistic Regression	97.20%	0.97	0.97	0.97
SVM (Linear)	97.85%	0.99	0.97	0.98

5.2 Analysis of Results

1. **SVM Superiority:** The SVM model achieved the highest accuracy (~97.8%). This confirms that finding the optimal margin is more effective than simple probability counting (Naive Bayes) for complex text data.
2. **High Baseline:** Even the simplest model, Naive Bayes, achieved >96% accuracy. This indicates that the vocabulary between "Sports" (team, score, win, season) and "Politics" (government, law, rights, israel) is highly distinct.
3. **Confusion Matrix Analysis:**
 - The Logistic Regression model misclassified only a handful of documents.
 - *False Positives:* Some political articles discussing "strategy" and "winning" were mistaken for sports.
 - *False Negatives:* Sports articles discussing "legal disputes" or "bans" were mistaken for politics.

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--- Naive Bayes Results ---
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Accuracy: 0.9556
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Classification Report:
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	precision	recall	f1-score	support
Politics	0.94	0.99	0.96	549
Sports	0.98	0.91	0.94	375
accuracy			0.96	924
macro avg	0.96	0.95	0.95	924
weighted avg	0.96	0.96	0.96	924

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Confusion Matrix:
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[[543  6]
 [ 35 340]]
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6. Limitations and Conclusion

6.1 System Limitations

- **Static Vocabulary:** The model is trained on a fixed vocabulary. If new slang or new political figures (names not in the training set) appear in future documents, the model may fail to recognize them.
- **Context Blindness:** By using a "Bag of Words" (TF-IDF) approach, the model ignores sentence structure. It cannot understand sarcasm or negation (e.g., "This was not a victory" might be classified as positive due to the word "victory").
- **Cross-Domain Ambiguity:** The classifier struggles with intersectional topics. For example, an article about "Legislation regarding stadium funding" contains strong signals from both classes, leading to lower confidence scores.

6.2 Conclusion

This project successfully demonstrated the effectiveness of traditional machine learning for text classification. While all three models performed exceptionally well, the **Support Vector Machine (SVM)** proved to be the most robust choice. For real-world deployment where inference speed is critical, **Logistic Regression** would be a strong alternative as it offers similar accuracy with faster prediction times than SVM. Future work could involve implementing Deep Learning models like BERT to better capture the semantic meaning and context of the documents.

7. References

1. Scikit-learn: Machine Learning in Python, Pedregosa et al., JMLR 12, pp. 2825-2830, 2011.
2. Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval*. Cambridge University Press.
3. The 20 Newsgroups Dataset: <https://www.kaggle.com/datasets/crawford/20-newsgroups>